

scripts/autonomous-qa/lib-emulator.sh — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)

Inheritance: HelixConstitution §11.4.18 (script docs); Lava §6.AG / §6.AH / §6.X (containerized emulators), §6.R (no-hardcoding, slirp exemption) **Classification:** project-specific

Overview

Sourceable library of thin Lava glue replicating submodules/
containers/pkg/emulator/containerized.go to boot the §6.X
containerized KVM Android emulator image and wire host adb. The
emulator process runs INSIDE a podman container (image built from
the submodule Containerfile), NEVER host-direct, NEVER a live device.
The image bundles the AVD default, an adb keypair, and a socat bridge
(container 0.0.0.0:5575 → 127.0.0.1:5555 adbd).

Functions

Function	Purpose
emu_pick_port	Pick a free TCP port on loopback
emu_boot [avd]	Boot the container (cold boot); prints the container name; persists state
emu_authorize_adb <container>	Copy the image's baked adb key out and point the host adb client at it (ADB_VENDOR_KEYS)
emu_connect <adb_port>	adb connect; prints the adb serial (127.0.0.1:<port>)
emu_wait_boot <serial> [timeout_secs]	Poll sys.boot_completed + pm ready (default 360 s); dumps container logs on timeout
emu_fix_network <serial>	Ensure the guest has an active default network (see below)

Function	Purpose
<code>emu_cleanup_orphans</code>	Remove leftover <code>lava-emu-*</code> containers from an interrupted run
<code>emu_teardown [container]</code>	Remove the container + the state file

State + configuration

- State file: `scripts/autonomous-qa/.emu-state (CONTAINER / ADB_PORT / CONSOLE_PORT)` — written by `emu_boot`, removed by `emu_teardown`.
- `EMU_IMAGE` (env, default `ghcr.io/vasic-digital/lava-android-emulator:api34-x86_64`) — the emulator image.
- Container run flags: `--usersns=keep-id --device /dev/kvm, host-loopback port publishing only, ANDROID_COLD_BOOT=true`.

`emu_fix_network` — the `eth0/ndc` default-network fix

On some hosts (observed: podman 4.9.3 on `thinker.local`) the emulator boots with `eth0` DOWN and Android never registers it as a framework network — apps see "Network unreachable" because Android routes per-network via `netd` mark-based tables. The function (root via `su 0` inside the guest) brings `eth0` up and adds it to the default physical network (`netId 100`) through `ndc`, retrying up to 8 times until the `slirp` gateway pings. Self-gating + idempotent: if the gateway already answers it no-ops; it never fails the run — the iteration records the REAL outcome (anti-bluff: a stalled onboarding shows as SKIP/FAIL).

§6.R note: the `10.0.2.x` literals are platform-FIXED Android-emulator QEMU user-mode (`slirp`) constants hardwired by the emulator — `10.0.2.2` = host-loopback alias, `10.0.2.3` = built-in DNS, `10.0.2.15` = the guest NIC. They are the emulator equivalent of `127.0.0.1` and cannot drift; the §6.R IPv4 scanner carries a narrow path+range exemption for the `10.0.2.0/24` `slirp` range scoped to this helper (see `scripts/scan-no-hardcoded-ipv4.sh`).

Usage

```
source scripts/autonomous-qa/lib-emulator.sh
emu_cleanup_orphans
CONTAINER="$(emu_boot default)"
```

```
ADB_PORT="$(grep '^ADB_PORT=' scripts/autonomous-qa/.emu-state |  
    cut -d= -f2)"  
emu_authorize_adb "$CONTAINER"  
SERIAL="$(emu_connect "$ADB_PORT")"  
emu_wait_boot "$SERIAL" 360  
emu_fix_network "$SERIAL"  
# ... run tests against $SERIAL ...  
emu_teardown
```

Companion files

- scripts/autonomous-qa/run-matrix.sh, run-nav-challenges.sh, release-coldstart-canary.sh — the consumers
- submodules/containers/pkg/emulator/ — the canonical Go implementation this mirrors
- docs/scripts/run-emulator-tests.sh.md — the sibling Containers-driven emulator runner